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Editor-in-Charge
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Dear Editor-in-Charge,

We thank you for your decision and for taking the time to explain it. We respectfully ask that you consider a substantially revised version of the manuscript, “Which quantum algorithms suit photonic continuous-variable hardware? A comparative resource and noise-robustness study,” attached here, before our final redirection to another venue. We accept your earlier judgement that the original submission did not contain a sufficiently concrete physics result for PRA; in the revision we have addressed that point directly with new physics content, summarised below.

What is new since the previous submission.

Worked physics example on three diatomic molecules (new Sec. VIII, new Fig. 4, new Table II). We compute the vibrational eigenstructure of H_2 , HF , and I_2 from their Morse potentials in second-quantised form, using experimentally fixed Huber–Herzberg spectroscopic constants (no free parameters). Control-free CV quantum phase estimation, applied to the resulting Morse Hamiltonians, recovers the populated vibrational eigenenergies of all three molecules to within 0.55 cm^{-1} , i.e. *below the 1 cm^{-1} spectroscopic threshold standard in molecular physics*. The recovery spans a decade in vibrational quantum ($\omega_e = 214\text{--}4401 \text{ cm}^{-1}$) and an order of magnitude in anharmonicity ($x_e = 0.003\text{--}0.028$); the weakly anharmonic case (I_2) reaches $\sim 10^{-2} \text{ cm}^{-1}$, roughly two orders of magnitude inside the spectroscopic threshold.

Quantitative resource statement for a real physical observable. On the same H_2 vibrational spectrum, reaching 0.5 cm^{-1} accuracy with discrete-variable QPE requires roughly 1.5×10^{10} T-gates and 1.6×10^4 controlled time evolutions; the CV control-free protocol needs *zero* controlled unitaries and *zero* non-Gaussian gates for the harmonic part. The advantage is not incremental; it is structural and quantitative on a *physically observable spectrum*.

Validation and reproducibility. The Hamiltonian construction is exact in the truncated Fock basis (verified eigenvalue agreement with the analytic Morse formula to machine precision at suitable cutoff), the recovery is performed by reference-eigenstate holographic phase retrieval, and every reported number is reproduced by a single self-contained script bundled with the submission.

Position relative to PRA scope. Computing the vibrational energy levels of real diatomic molecules to spectroscopic accuracy is, in our view, a physical observable squarely within PRA’s scope; the algorithmic infrastructure that surrounds it is the means by which that observable is computed. The three-molecule demonstration is not a single toy example: it spans a wide swath of the molecular vibrational parameter space. Beyond the methodology contribution of the original manuscript, the new section provides a quantitative resource prediction that experimentalists can directly test on near-term GKP-encoded photonic processors.

We are aware that the threshold for PRA is high and that the editor's discretion is final. We submit this revised version not to dispute the earlier judgement on the original manuscript, but to ask whether the addition of the $\text{H}_2/\text{HF}/\text{I}_2$ spectroscopic-accuracy calculation changes the assessment. If it does not, we will redirect the work to another venue without further appeal, and we are grateful for the careful consideration given so far.

The manuscript has not been published and is not under consideration elsewhere; the authors declare no competing interests. All scripts and numerical results that reproduce every figure and table are included with the submission.

Sincerely,

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QIRI (Quantum Integrated Research
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